

Module 5: Data Modeling Fundamentals

Overview

Data modeling is one of the most important concepts in Power BI. A well-designed data model improves report performance, simplifies DAX calculations, and ensures accurate business insights.

Many Power BI beginners focus heavily on visualizations and DAX but overlook data modeling. In reality, most performance and reporting issues originate from poor data models.

This module covers relationships, cardinality, filter propagation, keys, and best practices for designing scalable Power BI models.

Learning Objectives

After completing this module, you will be able to:

- Understand Data Modeling concepts
 - Understand the role of tables in Power BI
 - Create relationships between tables
 - Understand Primary Keys and Foreign Keys
 - Understand Cardinality
 - Understand Cross Filter Direction
 - Manage Relationships
 - Design efficient Power BI data models
 - Troubleshoot relationship issues
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1 Introduction to Data Modeling

What is Data Modeling?

Data Modeling is the process of organizing data and defining relationships between tables.

It determines how data is connected and how Power BI retrieves information when building reports.

Simple Example

Imagine a company has two tables:

Customers Table

CustomerID	CustomerName
101	Rahul
102	Priya

Orders Table

OrderID	CustomerID	Sales
1	101	5000
2	102	3000

Power BI uses CustomerID to connect both tables.

This connection is called a relationship.

2 Why Data Modeling is Important

A proper model provides:

Faster Performance

Efficient relationships reduce query execution time.

Accurate Results

Ensures calculations return correct values.

Easier Reporting

Users can build reports more easily.

Better Scalability

Handles large datasets efficiently.

Without Proper Modeling

Problems include:

- Duplicate results
 - Incorrect totals
 - Slow reports
 - Complex DAX formulas
-

3 Tables in Power BI

Power BI models consist of tables.

Types of Tables

Fact Tables

Contain measurable business data.

Examples:

- Sales
 - Orders
 - Transactions
 - Revenue
-

Dimension Tables

Contain descriptive information.

Examples:

- Customer

- Product
- Employee
- Region
- Date

Example

Sales Fact Table

OrderID	ProductID	CustomerID	Sales
1	P101	C101	5000

Product Dimension

ProductID	ProductName
P101	Laptop



Understanding Relationships

Relationships connect tables.

They allow Power BI to combine information from multiple tables.

Example

Customers

CustomerID	Name
101	Rahul

Orders

OrderID	CustomerID
1	101

Relationship:

```
Customers[CustomerID]
      ↓
Orders[CustomerID]
```

5 Primary Keys

Definition

A Primary Key uniquely identifies each row in a table.

Characteristics

- Unique
 - No duplicates
 - No null values
-

Example

Customers Table

CustomerID	Name
101	Rahul
102	Priya

CustomerID is the Primary Key.

Invalid Example

CustomerID
101
101

Duplicates make it invalid.

6 Foreign Keys

Definition

A Foreign Key is a column that references a Primary Key from another table.

Example

Customers Table

CustomerID

101

Orders Table

OrderID CustomerID

1	101
---	-----

CustomerID in Orders is a Foreign Key.

Purpose

Foreign Keys create relationships between tables.

7 Relationship Types

Power BI supports several relationship types.

One-to-One (1:1)

One record matches one record.

Example

EmployeeID

1

↔

EmployeeID

1

One-to-Many (1:*)

Most common relationship.

Example

One customer can place many orders.

Customer



Orders

Customers

CustomerID

101

Orders

OrderID CustomerID

1 101

2 101

Many-to-One (*:1)

Reverse view of One-to-Many.

Many-to-Many (:)

Both tables contain duplicates.

Example

Students and Courses

A student can enroll in many courses.

A course can have many students.

Cardinality

What is Cardinality?

Cardinality defines how tables relate to one another.

Types

One-to-One

1 : 1

One-to-Many

1 : *

Many-to-One

* : 1

Many-to-Many

* : *

Why Cardinality Matters

It affects:

- Data Accuracy
- Filter Propagation
- Query Performance



Cross Filter Direction

What is Cross Filtering?

Determines how filters move between tables.

Single Direction

Dimension
↓
Fact

Most recommended.

Example

Product filters Sales.

Sales does not filter Product.

Both Direction

Dimension ↔ Fact

Filters move in both directions.

Advantages

Useful for advanced scenarios.

Disadvantages

May:

- Reduce performance
 - Create ambiguity
 - Produce unexpected results
-

Best Practice

Use Single Direction whenever possible.

10 Active vs Inactive Relationships

Power BI allows multiple relationships between tables.

However, only one relationship can be active at a time.

Active Relationship

Represented by:

Solid Line

Used automatically.

Inactive Relationship

Represented by:

Dashed Line

Requires DAX functions like:

USERELATIONSHIP()

Example

Orders Table:

- Order Date
- Delivery Date

Date Table can only have one active relationship.

Creating Relationships

Automatic Detection

Power BI attempts to detect relationships automatically.

Manual Creation

Method 1

Drag column to matching column.

Method 2

```
Model View
  ↓
Manage Relationships
  ↓
New
```

Required Information

- Table
 - Column
 - Cardinality
 - Filter Direction
-

1 2 Managing Relationships

Manage Relationships Window

Allows:

- Edit Relationships
 - Delete Relationships
 - Create Relationships
 - Activate Relationships
-

Relationship Settings

Cardinality

Choose relationship type.

Filter Direction

Single or Both.

Active Status

Enable or disable relationship.

1 3 Relationship Troubleshooting

Problem 1

Missing Relationship

Symptoms

Blank visuals.

Solution

Create relationship.

Problem 2

Duplicate Keys

Symptoms

Relationship creation fails.

Solution

Remove duplicates.

Problem 3

Incorrect Data Types

Symptoms

Relationship unavailable.

Solution

Ensure matching data types.

Problem 4

Ambiguous Paths

Symptoms

Incorrect filtering.

Solution

Simplify model.



Data Modeling Best Practices

Use Star Schema

Preferred structure.

Avoid Many-to-Many

Use only when necessary.

Use Meaningful Names

Examples:

```
FactSales
DimCustomer
DimProduct
```

Remove Unused Columns

Improves performance.

Create Date Table

Required for Time Intelligence.

Use Single Direction Filtering

Unless a specific need exists.

1 5 Common Modeling Mistakes

Loading Everything

Imports unnecessary data.

Using Flat Files Only

Creates inefficient models.

Ignoring Relationships

Causes reporting errors.

Excessive Bi-Directional Filtering

Reduces performance.

Poor Naming Conventions

Makes maintenance difficult.



Key Terminologies

Term	Definition
Data Model	Collection of connected tables
Relationship	Connection between tables
Primary Key	Unique identifier
Foreign Key	References another table
Cardinality	Relationship structure
Fact Table	Contains measurable data
Dimension Table	Contains descriptive data
Filter Propagation	Movement of filters
Active Relationship	Default relationship
Inactive Relationship	Secondary relationship



Practice Exercises

Exercise 1

Create two tables:

- Customers
- Orders

Build a relationship using CustomerID.

Exercise 2

Identify Primary Keys and Foreign Keys.

Exercise 3

Create:

- One-to-One Relationship
 - One-to-Many Relationship
-

Exercise 4

Experiment with:

- Single Filter Direction
- Both Filter Direction

Observe results.

Exercise 5

Create a model using:

- Product Table
 - Sales Table
 - Customer Table
-



Mini Project

Retail Sales Data Model

Tables

Customers

CustomerID	CustomerName
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Products

ProductID	ProductName
-----------	-------------

Sales

SaleID	CustomerID	ProductID	Sales
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Tasks

1. Import all tables
 2. Create relationships
 3. Identify Primary Keys
 4. Identify Foreign Keys
 5. Verify filter behavior
 6. Create basic report
-



Module Summary

In this module, you learned:

- ✓ Data Modeling Fundamentals
- ✓ Fact Tables & Dimension Tables

- ✓ Relationships
- ✓ Primary Keys
- ✓ Foreign Keys
- ✓ Relationship Types
- ✓ Cardinality
- ✓ Cross Filter Direction
- ✓ Active & Inactive Relationships
- ✓ Relationship Management
- ✓ Data Modeling Best Practices

A strong data model is the foundation of every successful Power BI solution. Before creating advanced visualizations and DAX measures, always ensure your model is designed correctly.



Next Module

Module 6: Star Schema & Snowflake Schema

Topics Covered:

- Dimensional Modeling
- Star Schema Design
- Snowflake Schema Design
- Fact & Dimension Tables
- Surrogate Keys
- Conformed Dimensions
- Performance Optimization
- Real-World Data Warehouse Models